ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 16:

Millimeter Wave Frequency Generation

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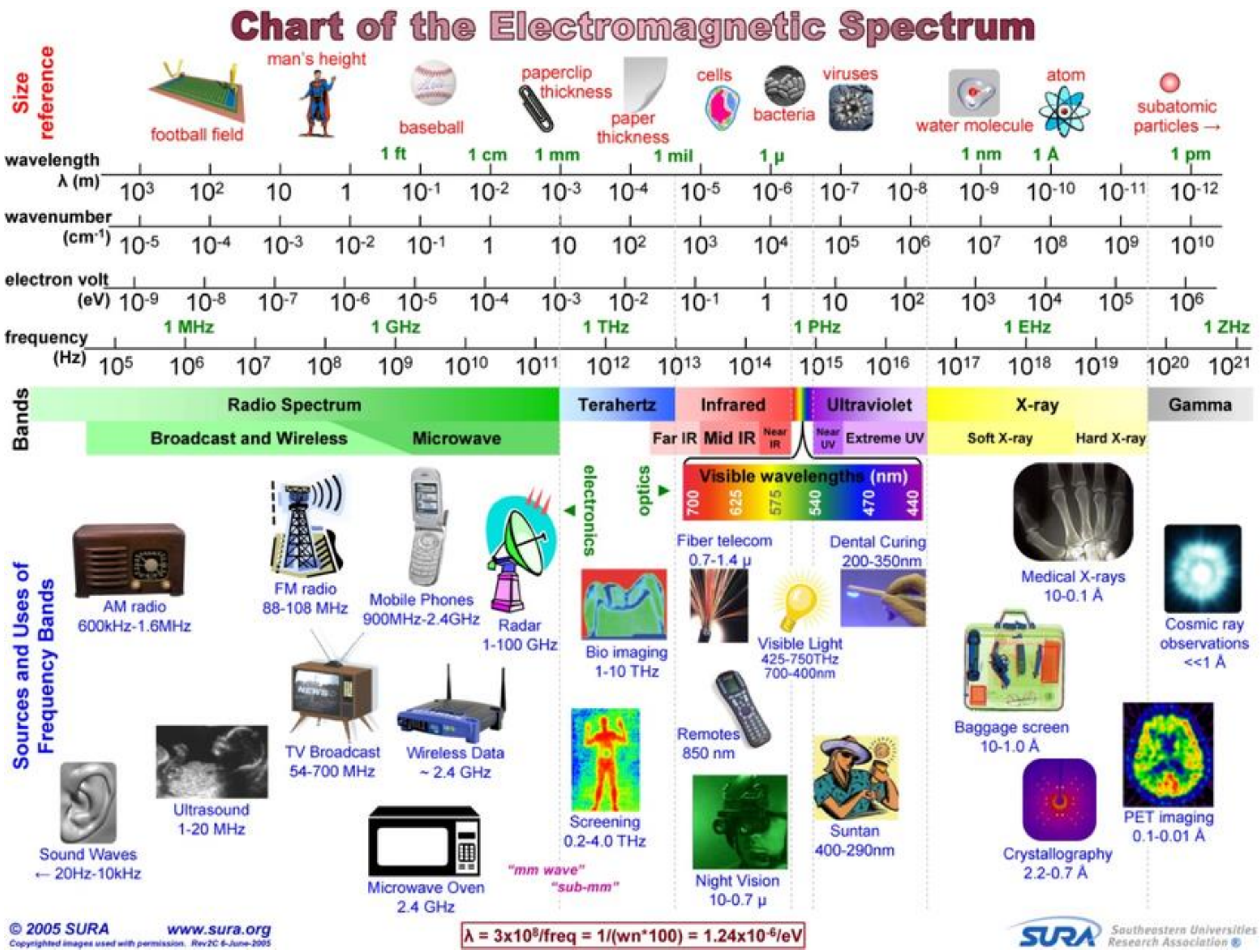
Outline

- Why millimeter wave?
- How is millimeter wave transceiver different from sub-6GHz transceiver?
- Challenges in millimeter wave frequency generation
- Key topics in millimeter wave frequency generation
  - Millimeter oscillators
  - Frequency multiplication
  - Phase shifting
  - High speed frequency divider
  - FMCW (frequency modulated continuous wave)
  - Improve phase noise performance at millimeter wave frequency from architecture level

Why Millimeter Wave?



[Ref] SURA Southeastern Universities Research Association

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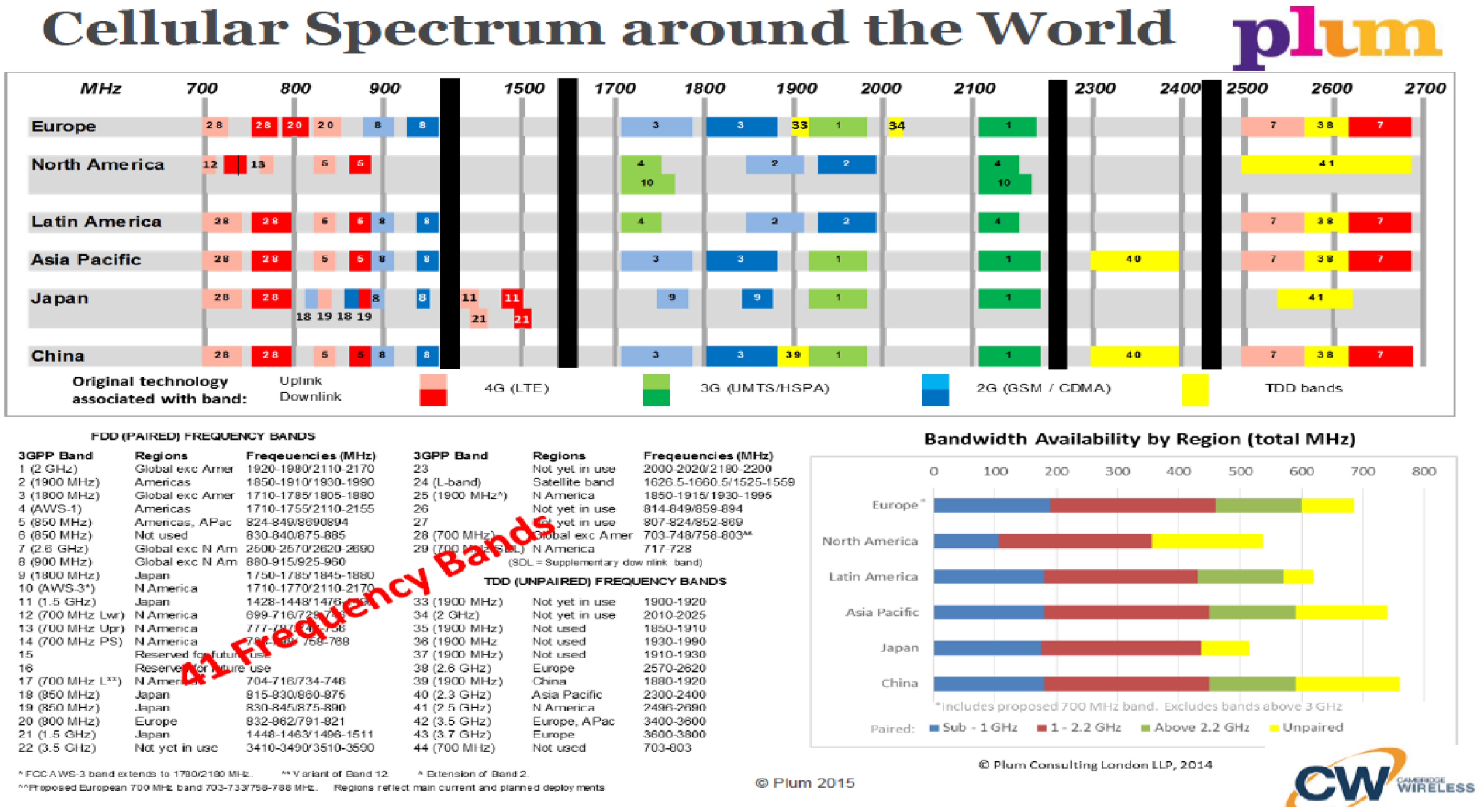
Why Millimeter Wave?



[Ref] Cambridge Wireless

- 40+ bands in sub-3GHz spectrum around the world
- Each band offers 20~100MHz bandwidth shared among multiple operators
- Crowded spectrum, various interference, limited data bandwidth and throughput

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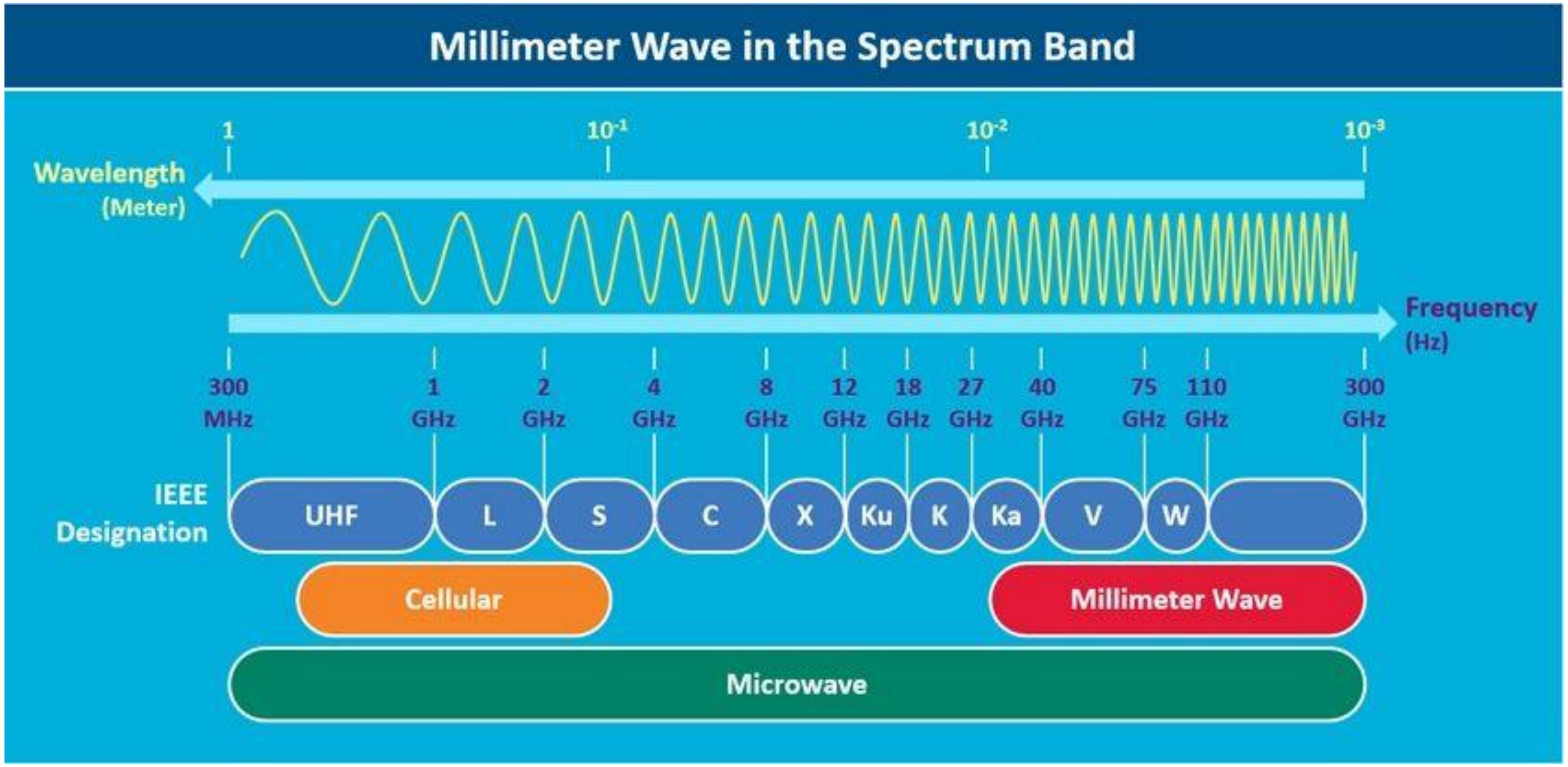
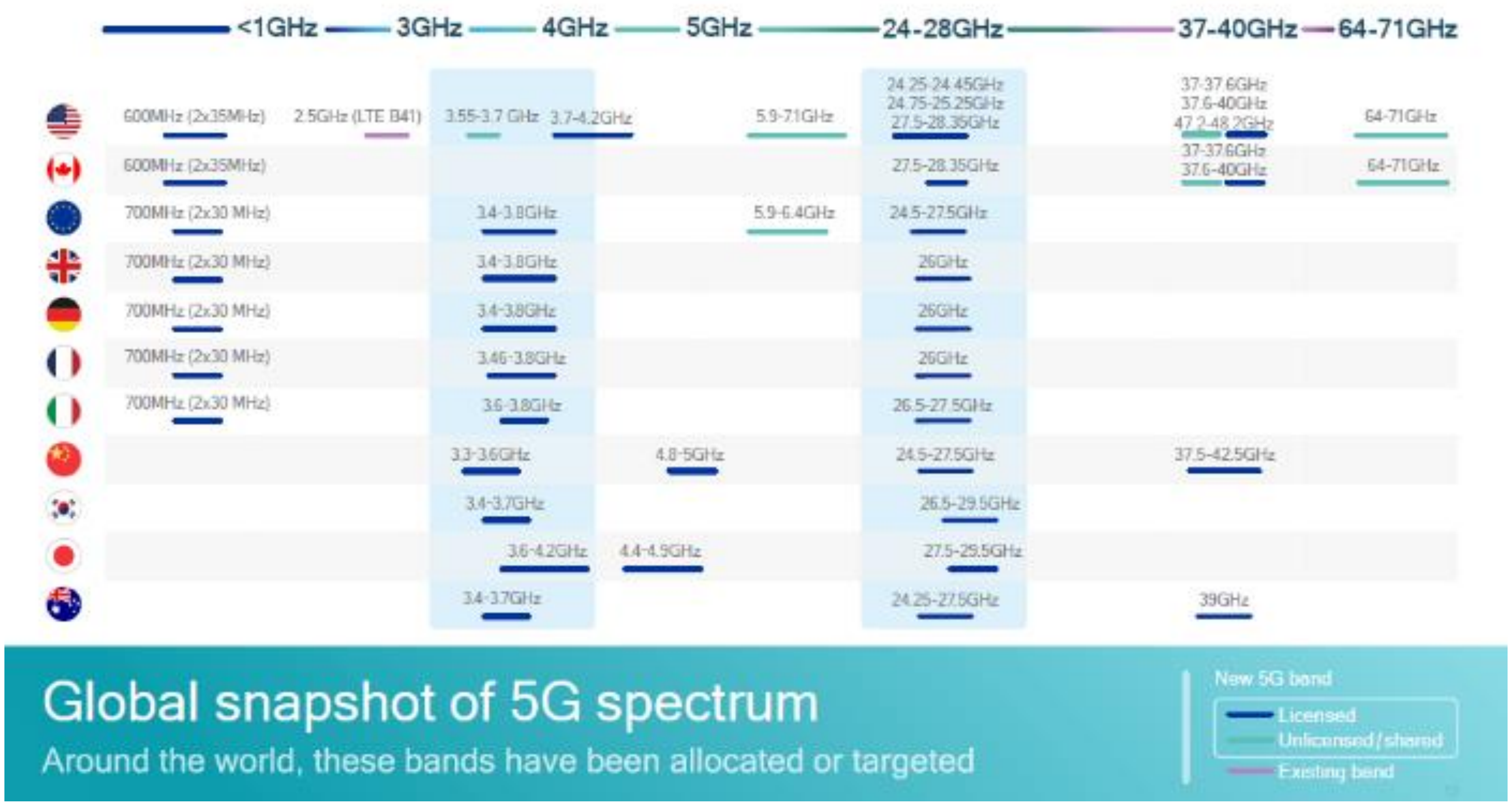
Millimeter Wave Spectrum



- Millimeter wave spectrum: 30~300GHz (λ = 10~1mm)
- Higher carrier frequency and wider bandwidth

[Ref] everythingrf.com/community/what-are-millimeter-waves

5G Cellular Spectrum

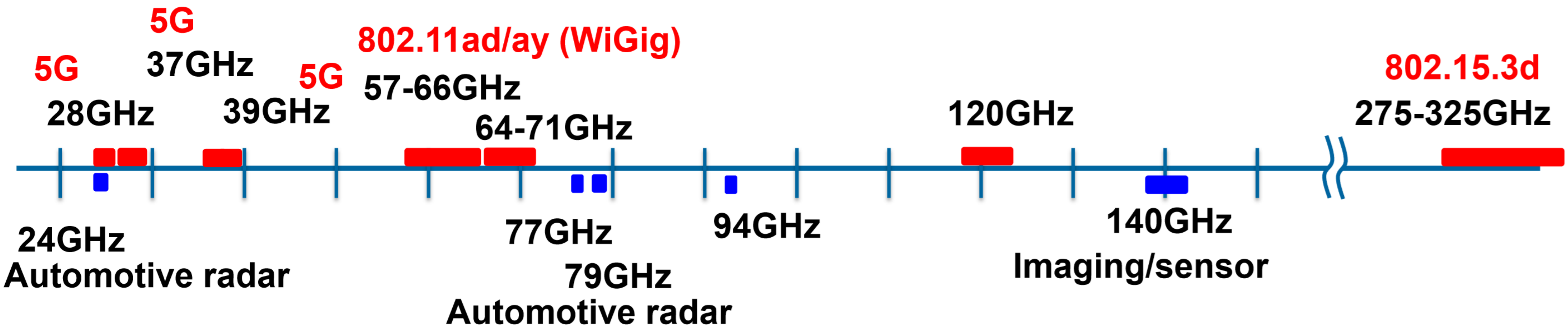


[Ref] everythingrf.com/community/5g-frequency-bands

- In 5G bands, mmW bands 24-28GHz and 37-48GHz bands are being deployed
- Each band provides ~GHz bandwidth for high data throughput
- More mmW bands will be added

Millimeter Wave Spectrum

Ref [1]



- Higher carrier frequency and wider bandwidth
- Applications
  - Communications (5G cellular, WLAN, 100Gbps point to point backhaul)
  - Automotive radar
  - Imaging, sensor, spectroscopy

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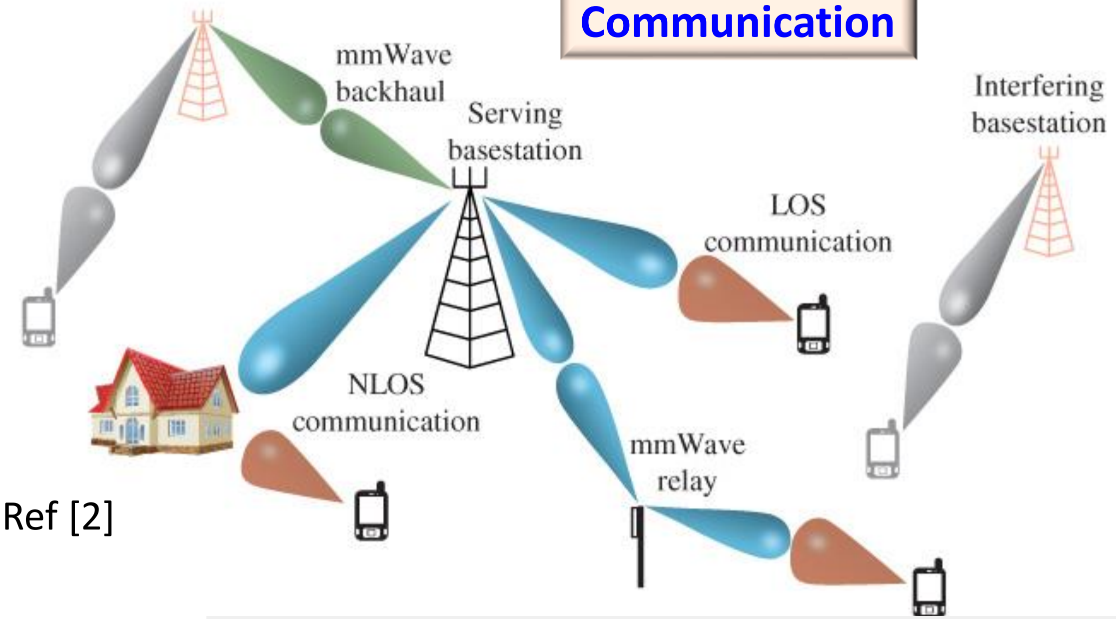
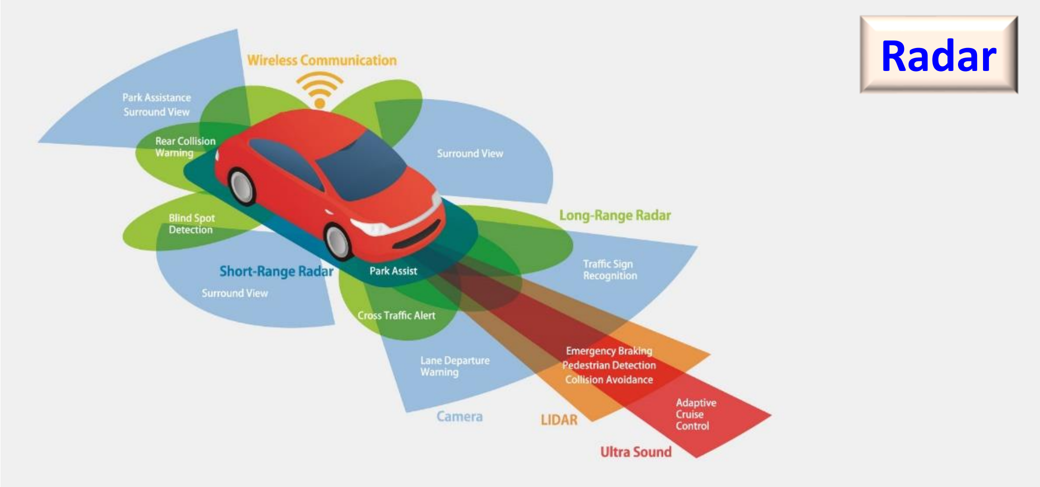
Millimeter Wave Applications

Communication

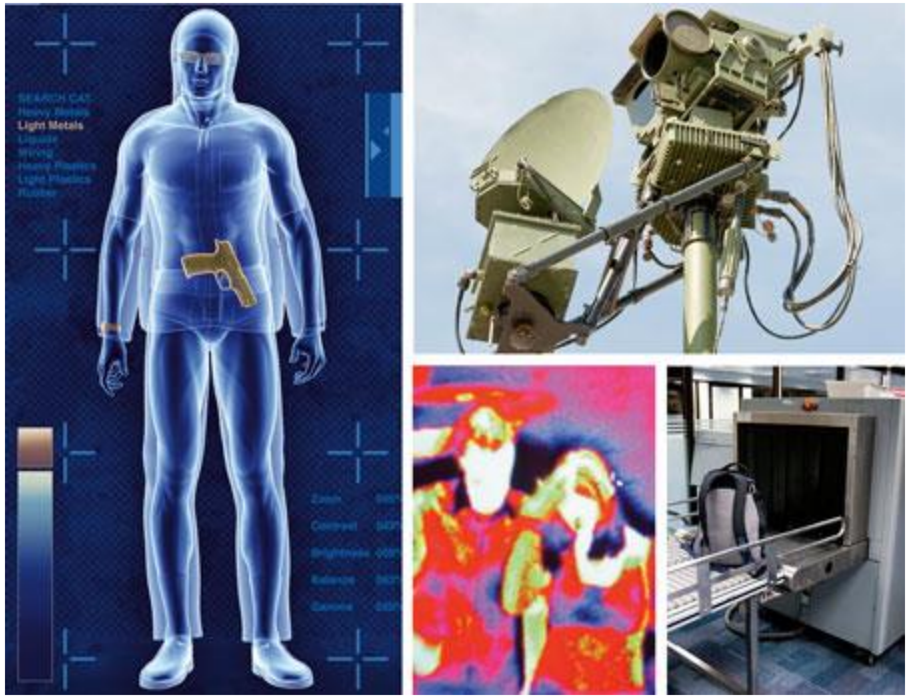


Ref [2]

Radar



[Ref] masttechnologies.com/automotive/



[Ref] Millimeter waves: emerging markets, Thintri.com

Imaging

How is Millimeter Wave Different?

	Millimeter Wave	Conventional RF bands
Frequency f	30 ~ 300GHz	400MHz ~ 6GHz
Wavelength λ	1 ~ 10mm	5 ~ 75cm
Path loss $L = 20log_{10}\left(\frac{4\pi d}{\lambda}\right)$	High	Low
Antenna size (~ λ/2)	Small	Large
Antenna array	Feasible for UE	Not feasible for UE
Antenna gain	High	Low
Signal propagation	Directional	Omnidirectional
Spatial division multiplexing	Yes	No
Polarization diversity	Yes	No
IC design challenge	Closer to f_{max}, higher routing loss, more sensitive to parasitic, higher power consumption	

Millimeter Wave Propagation

Line of Sight (LOS) Propagation

- Free space propagation without considering the following
  - Atmospheric, rain and foliage attenuation
  - Building penetration loss, user hand loss
  - Scattering effects and diffraction
- Friis' free space path loss equation

$$L = 20log_{10}\left(\frac{4\pi d}{\lambda}\right) = 20log_{10}\left(\frac{4\pi d}{c}f\right)$$

  - The longer the distance the higher the loss
  - The higher the frequency the higher the loss
- Use 2GHz as baseline, for the same distance
  - 24GHz, 39GHz and 60GHz has **22dB, 26dB and 30dB** more loss, respectively

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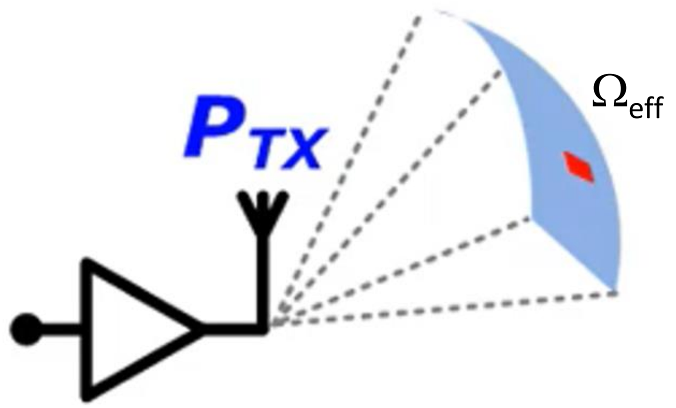
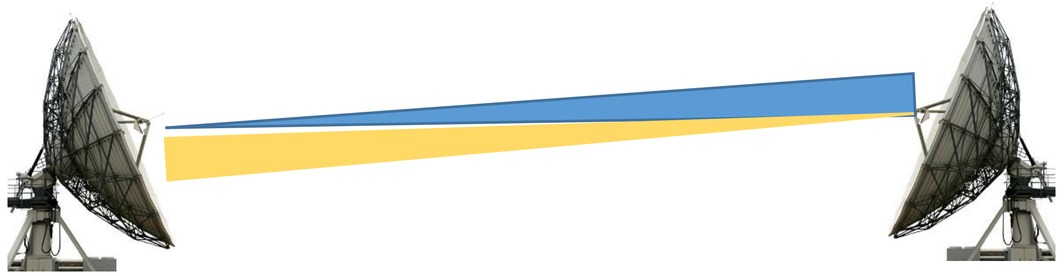
Isotropic vs. Directional Radiation

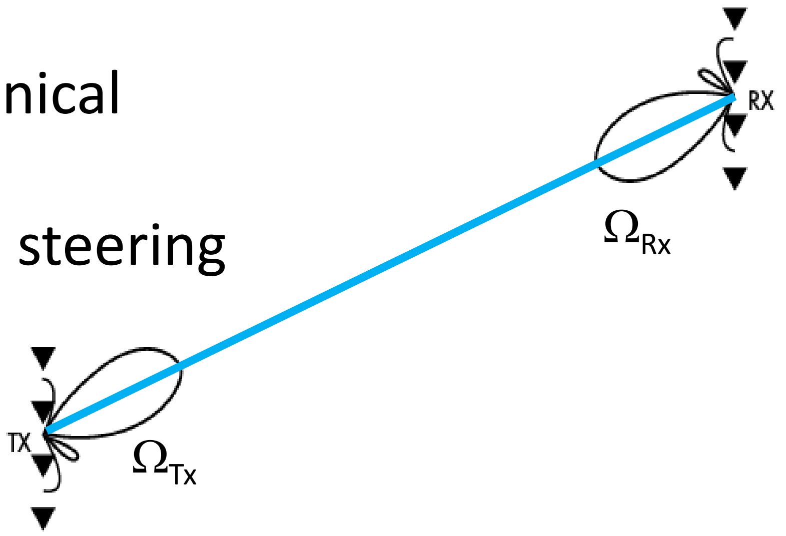
- Isotropic (omnidirectional) radiation
  - No alignment is needed, all directions are covered
  - Most of energy is wasted
  - At one frequency, one communication link at a time
  - More susceptible to multi-path fading
  - More interference
- Directional radiation
  - Alignment is needed (direction and polarization)
  - More energy efficient
  - At one frequency, multiple communication links at a time (spatial frequency reuse)
  - Less susceptible to multi-path fading, less interference

Directional Radiation – Antenna Gain

- Omnidirectional radiation solid radiation angle 4π
- Directional radiation beamwidth solid angle Ω_{eff}
- Antenna gain $G=4\pi/\Omega_{eff}$
- Antenna reciprocality: $G_{Rx} = G_{Tx}$
- How to achieve directional transmit and receive?
  - Concave reflective antenna for beamforming and mechanical beaming alignment
  - Antenna phased array beamforming and electrical beam steering

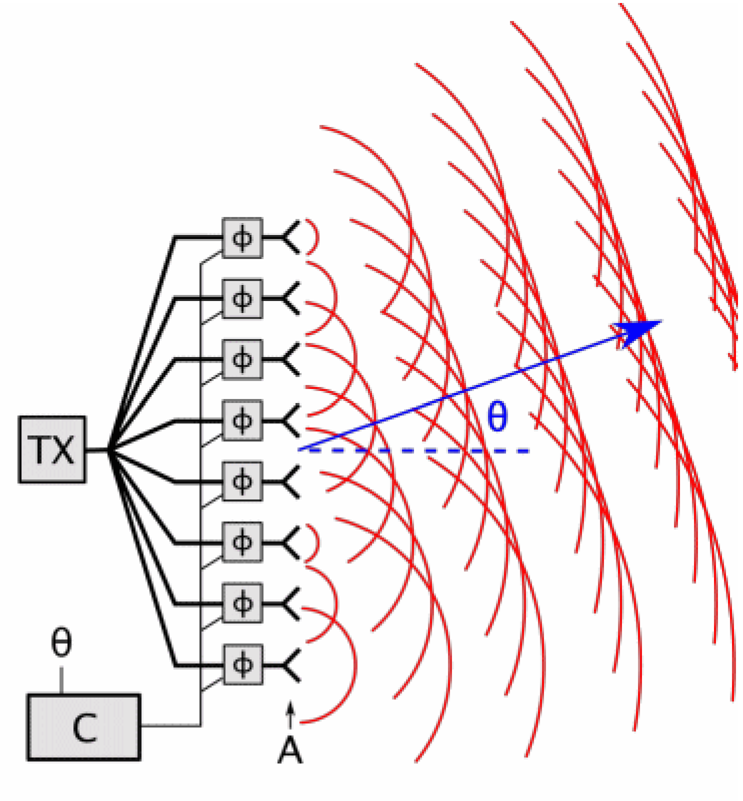






Phased Array and Beam Forming

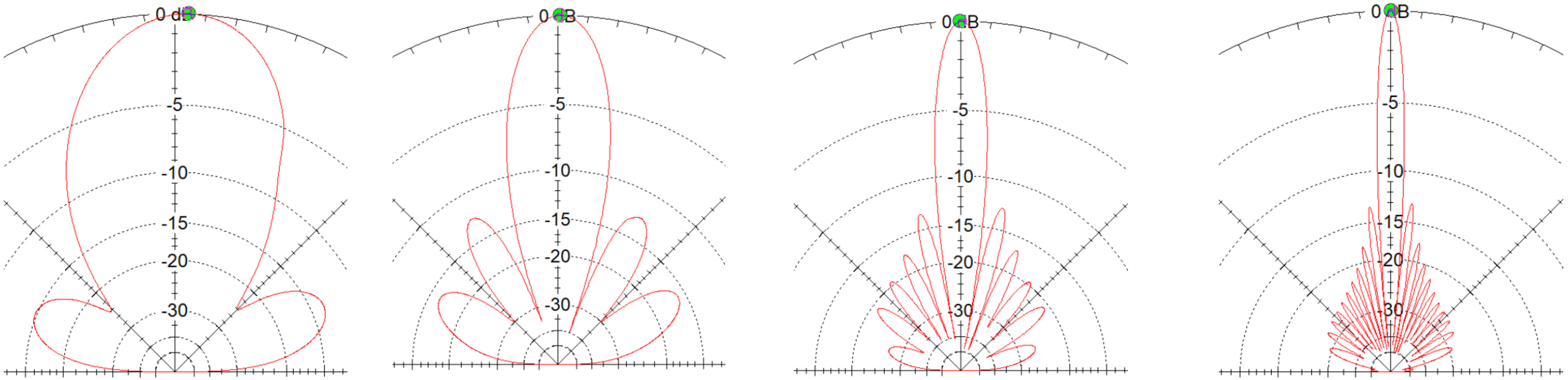
- Phased array
  - An assembly of radiating elements (antennae) in an electrical and geometrical configuration.
  - Total field of the array is determined by the vector addition of the field of each individual element
  - To generate directive patterns, the fields from the elements interfere constructively in the desired direction and interfere destructively in the remaining space.
  - A radiation beam is formed in the constructive direction with certain beam shape and beamwidth. Radiation energy is more directive and the communication link is more efficient.



Beam Forming and Beam Steering

- Beam forming
  - More antenna elements generate higher directive beams with narrower beamwidth
  - The higher the directivity, the higher the antenna array gain

Linear dipole array beam pattern



2 elements 4 elements 8 elements 16 elements

Beam Forming and Beam Steering

- Beam pattern can be changed by
  - Geometrical configuration of the overall array (linear, rectangular, circular, etc.)
  - Relative displacement between the elements ($\lambda/2$, $\lambda/4$, etc.)
  - Excitation amplitude of the individual elements
  - Excitation phase of the individual elements
  - Radiation pattern of the individual elements
- Beam direction is most effectively controlled by the excitation phase of individual elements
- Beam forming and beam steering are the key techniques to allow spatial division multiplexing and directional interference cancellation

Why Millimeter Wave?

- Antenna size
  - Proportional to wavelength
  - Antenna dimension $\lambda/2$ and array spacing $\lambda/2$
    - At 2.4GHz, $\lambda/2$ ~6cm
    - At 39GHz, $\lambda/2$ ~4mm
  - 5G antenna module
    - Qualcomm QTM052 supports 26~30GHz, 37~40GHz bands
    - Antenna module fits in the side of a UE device

2.4GHz WLAN antenna array

[Ref] 5GIS 2019

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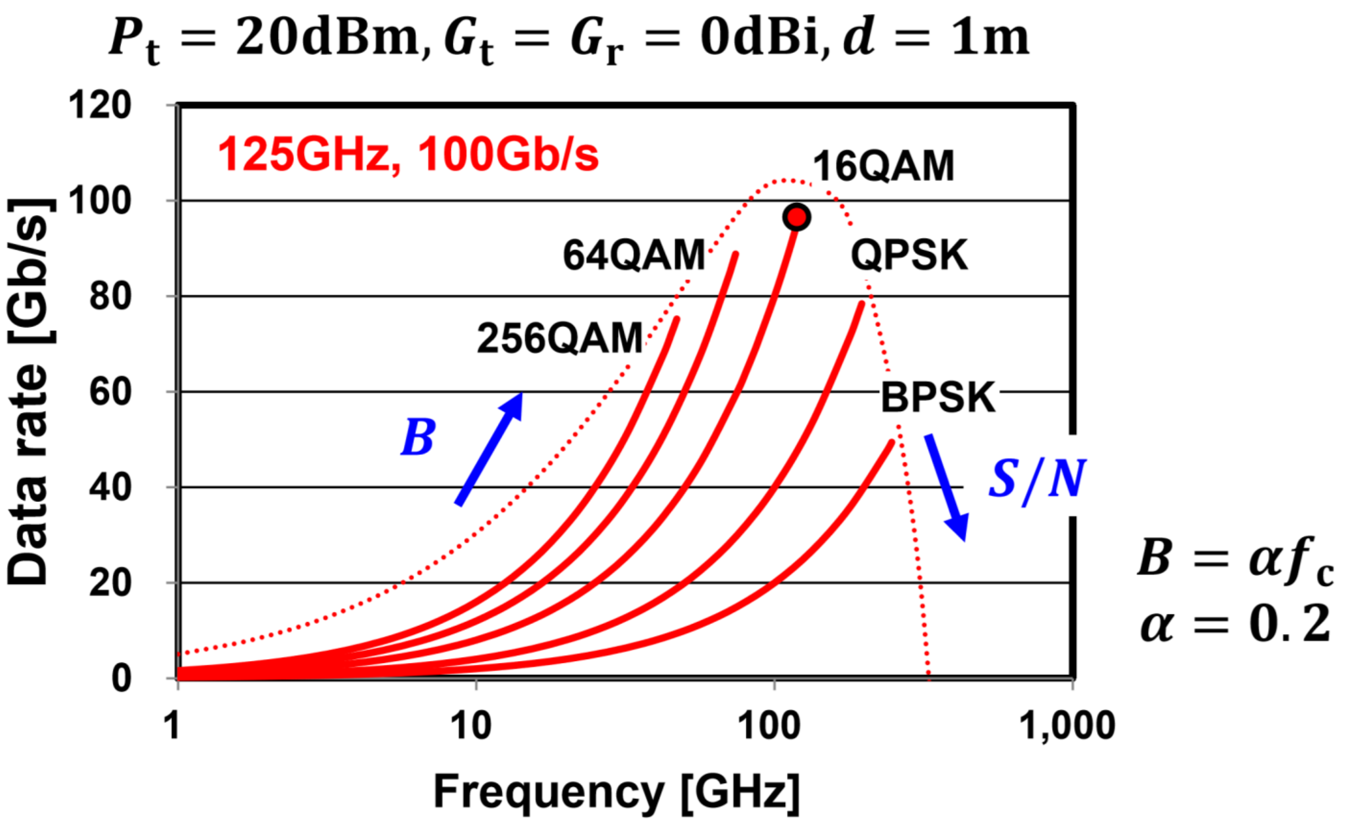
Design Challenges

- Link budget at millimeter wave frequency
  - High path loss and sensitive to environment
  - High directivity
- Silicon devices operating close to f_T/f_{max}
  - Limited available gain
  - Distributed effect within device
- Modeling accuracy and simulation to silicon correlation
  - Device model at high frequency
  - Interconnect and layout parasitic modeling
- High integration of large phased array
  - Coupling and isolation
  - Thermal density
- Characterization and testing
  - Conducting board characterization
  - Over the air characterization

The Higher the Frequency the Better?

- Communication applications
  - Data rate and power consumption considerations (bit/J)
  - Current cost effective technology (CMOS, SOI, SiGe) has a f_{max} of 200~500GHz.
  - As the operating frequency approaches f_{max},
    - Noise performance degrades: lower SNR lower data rate
    - DC to carrier frequency power conversion efficiency reduces: higher power consumption
  - Sweet spot frequencies



Ref [3] K. Okada, IEDM 2013

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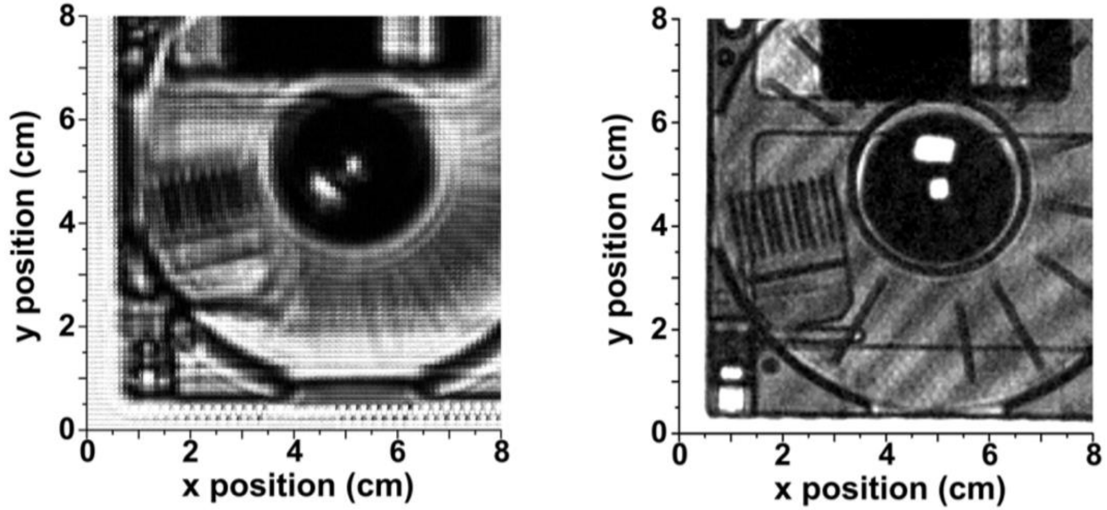
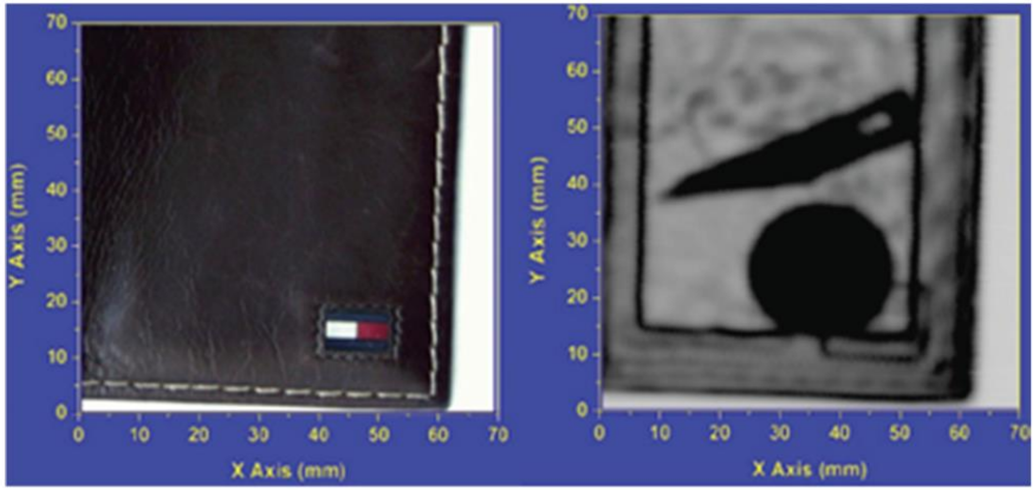
The Higher the Frequency the Better?

- Imaging applications:
  - Higher frequency provides resolution and penetration advantage.
  - At higher frequency, the wavelength is smaller, and bandwidth is larger. The image resolution is better. Above certain frequency, the radiation can penetrate to show the metallic objects.



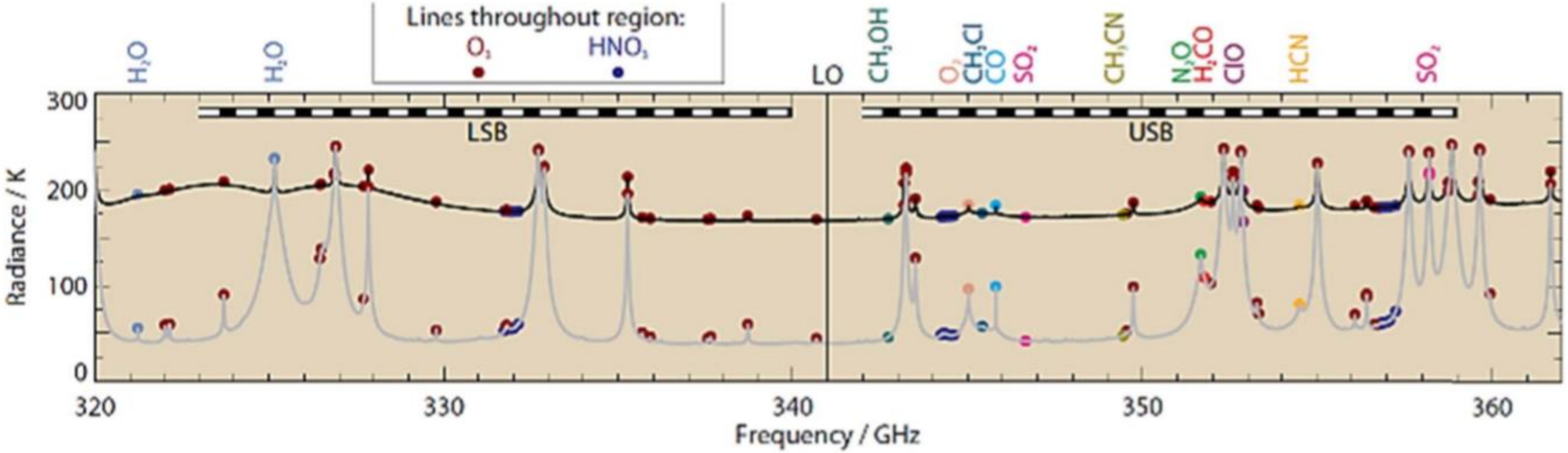
280GHz

860GHz



The Higher the Frequency the Better?

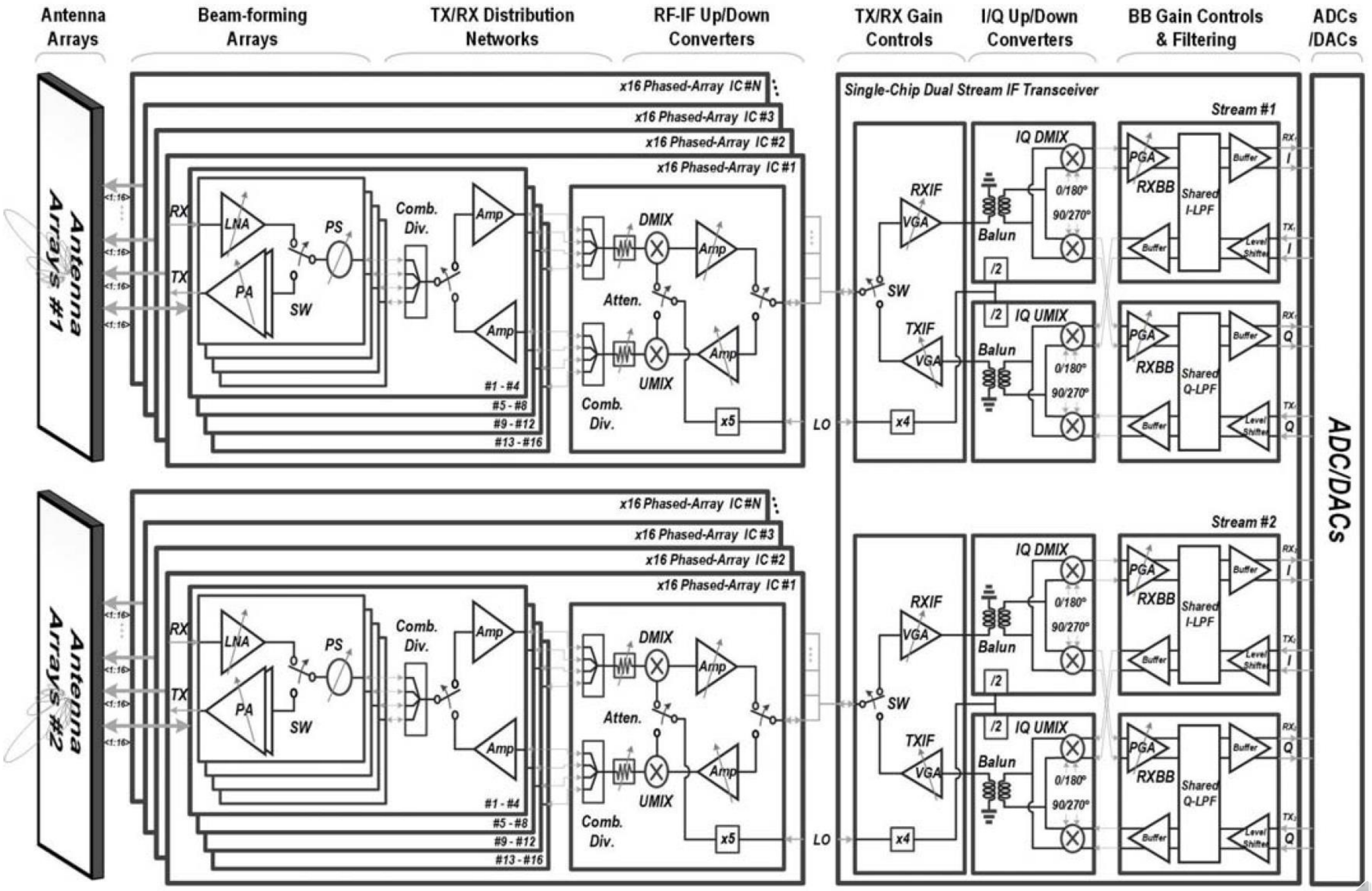
- Spectroscopy/spectrometer applications
  - Depending on the materials to be detected and their molecular spectral emission/absorption lines, the right frequencies are needed. For fast detection of multiple signature lines, fast frequency sweeping or wideband signal generation is needed



Ref [4]

Millimeter Wave Transceiver

- mm-W transceiver example
  - Transceiver array
  - Heterodyne
  - RF beamforming
  - Phase shifting in signal path



Ref [5]

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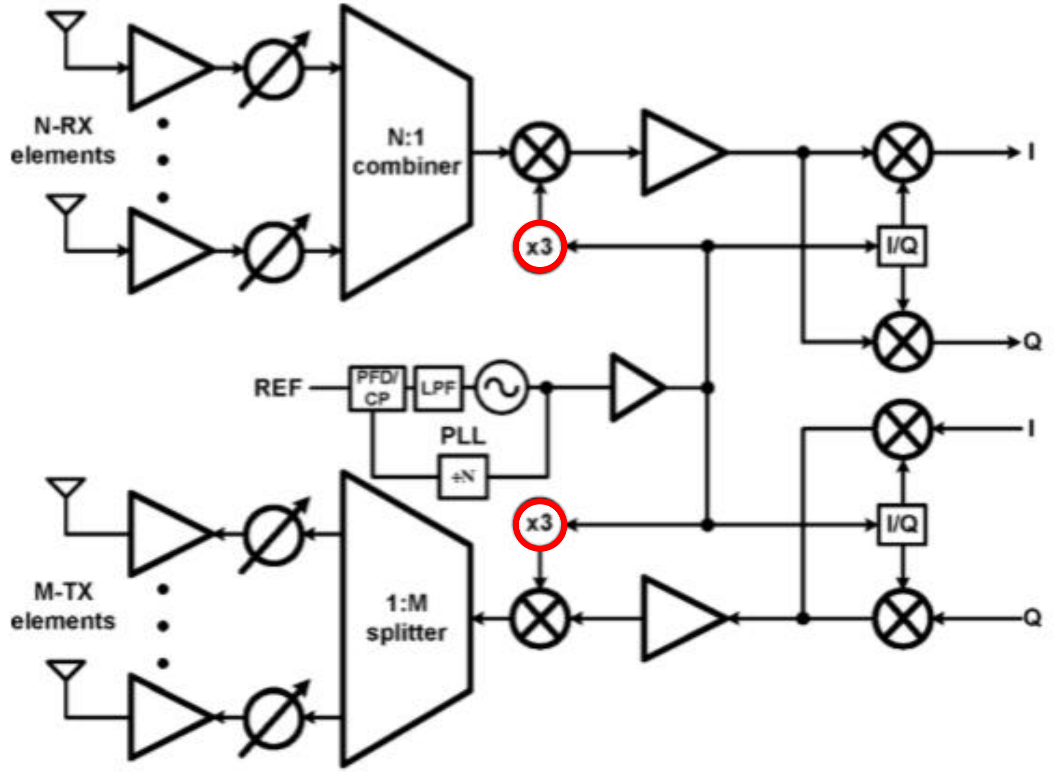
Millimeter Wave Frequency Generation

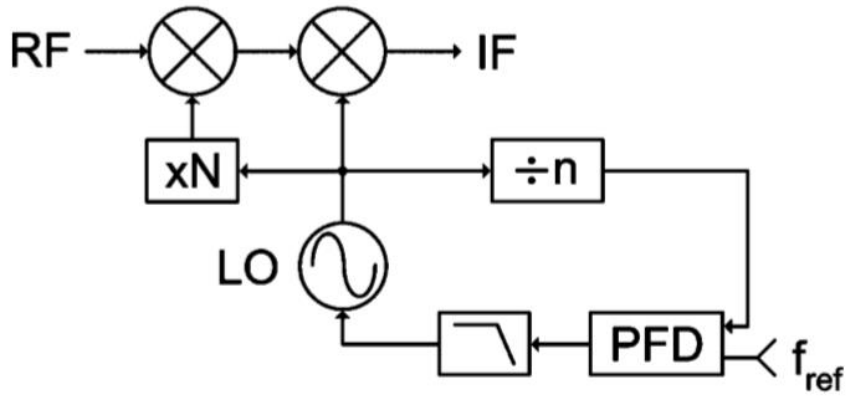
- Millimeter oscillators
- Frequency multiplication
- Phase shifting
- High speed frequency divider
- Improve phase noise performance at millimeter wave frequency from architecture level

Frequency Multiplication

A 120GHz Transceiver Array

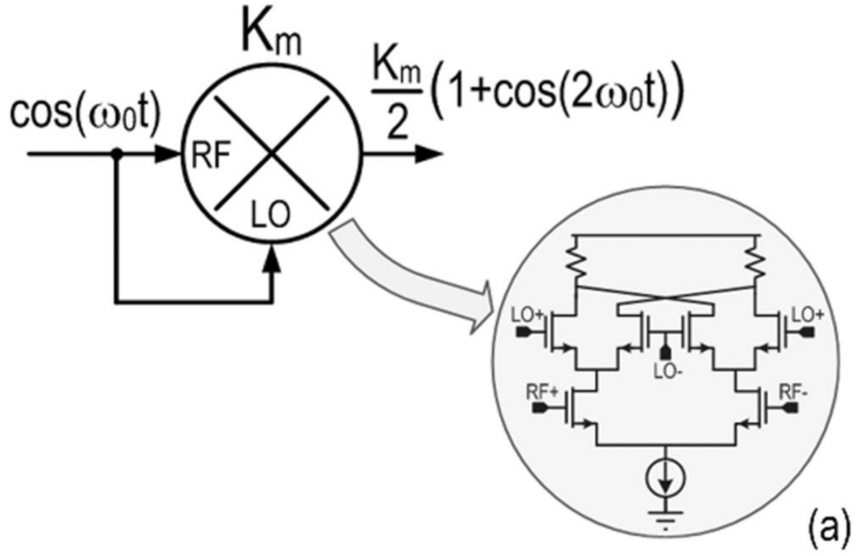


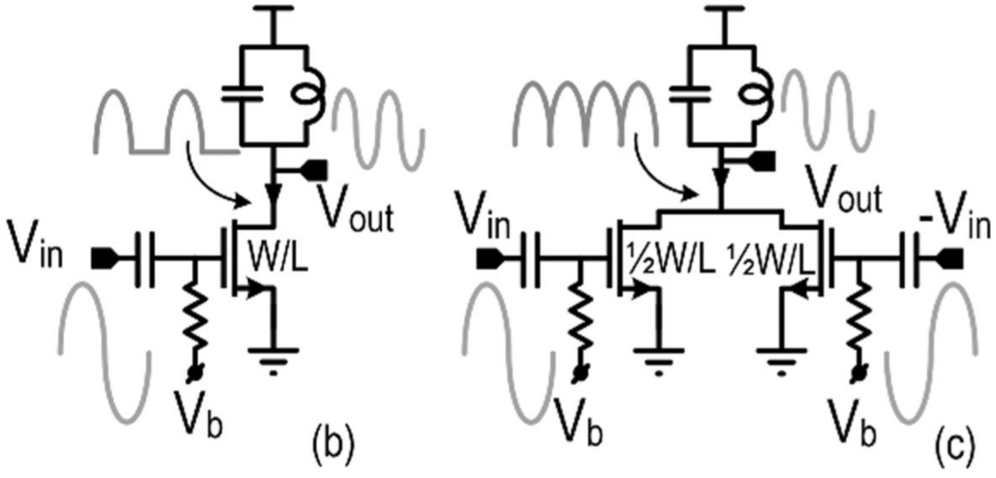
Ref [6]



- VCO figure of merit degrades significantly at mmW frequency
- VCO oscillates at lower frequency (30GHz)
- Tripler generates higher frequency LO signal (90GHz) for RF mixer
- I/Q generator (30GHz) drives IF mixer
- RF channel frequency at 120GHz

Frequency Doublers





Ref [7]

(a) Mixer based frequency doubler.
- The RF and LO ports of a Gilbert mixer are driven by the same input signal.
- Generating an output component at twice the input frequency
- Broad frequency range of operation

(b) Device nonlinearity based doubler
- MOS transistor is driven into compression
- Drain current is rich of harmonic components of the input signal.
- A resonant load selects the desired one (e.g., 2nd harmonic for frequency doublers) and suppresses the fundamental.
- For maximum conversion efficiency the device conduction angle must be low (class-B or class-C operation)

(c) Push-push frequency doubler
- Differential pair and differential input
- Effectively suppress the fundamental and odd harmonics

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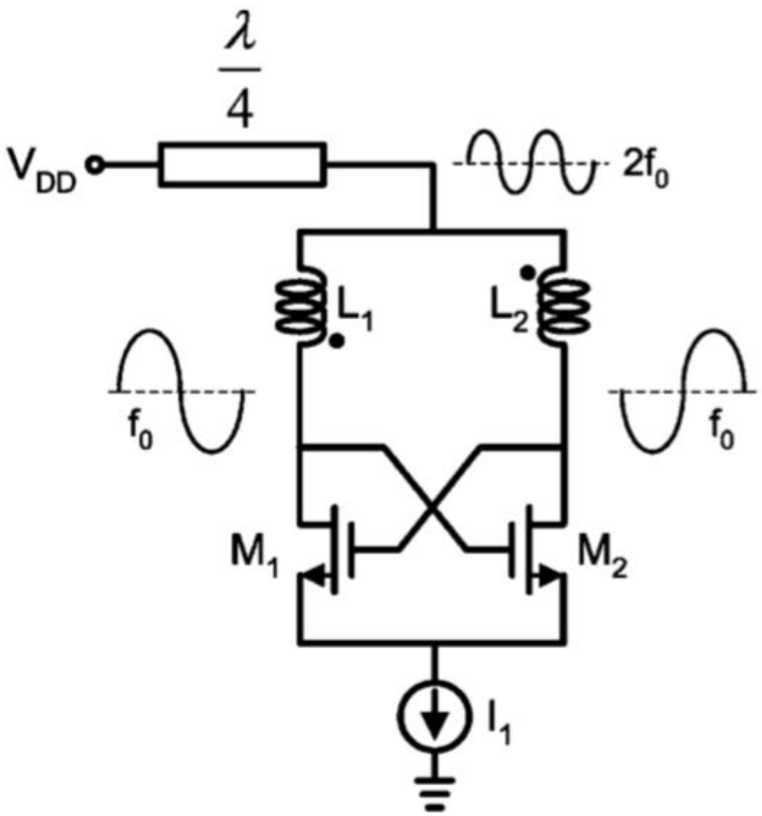
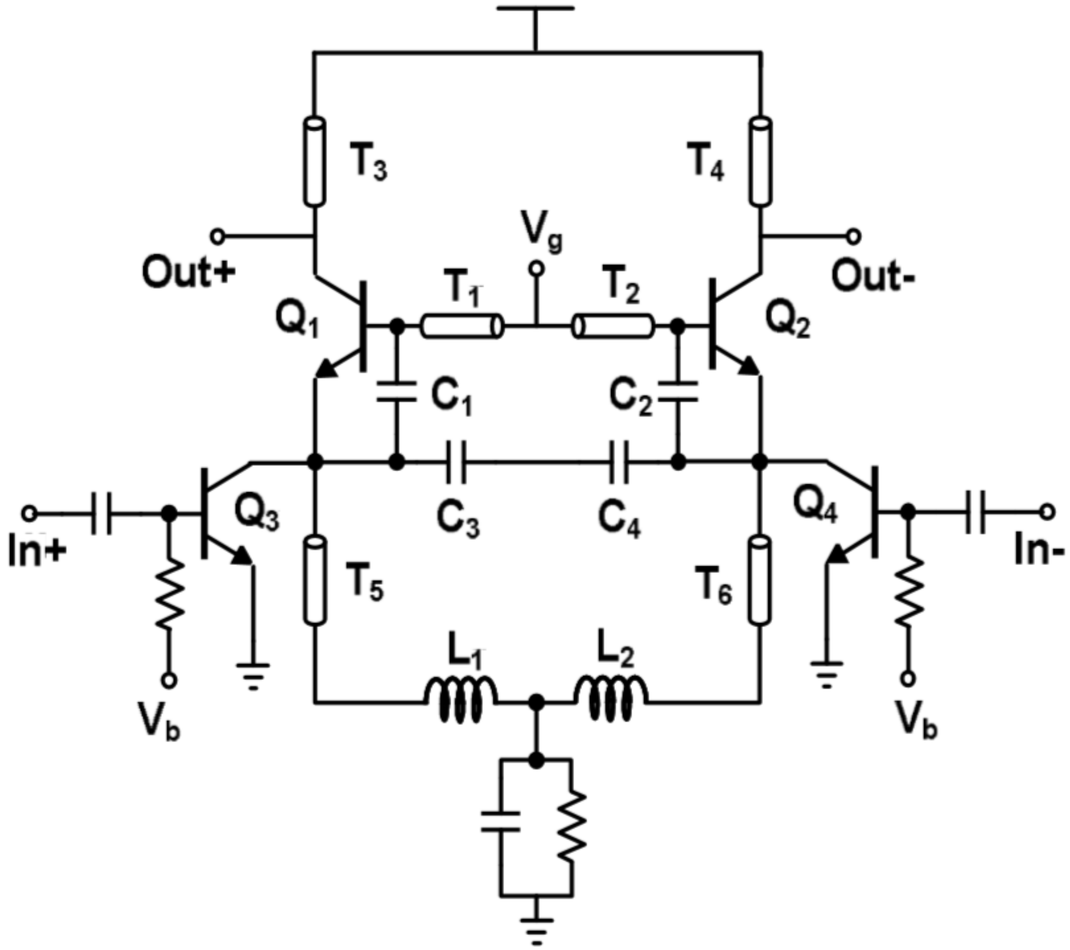
Push-Push Oscillator



- A strong second-harmonic tone appears at the virtual ground nodes in differential oscillator topologies.
- A quarter wavelength ($\lambda/4$) transmission line tuned for the second harmonic frequency is usually used to increase the amplitude of second harmonic while suppressing the fundamental signal.

$$\boxed{Z_{in}(\lambda/4) = \frac{Z_0^2}{Z_L}}$$

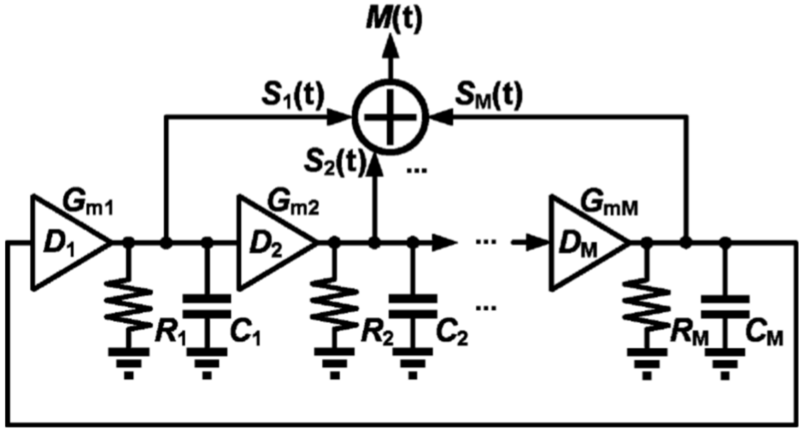
Frequency Tripler



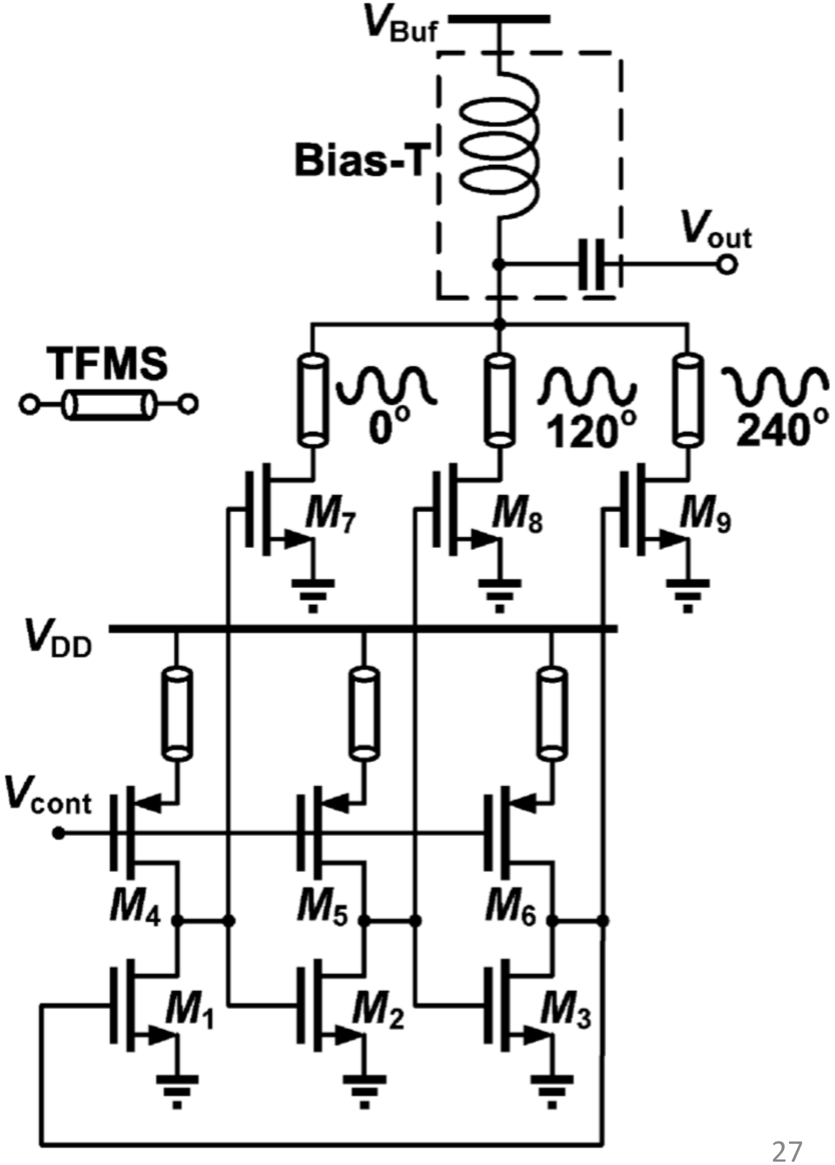
Ref [8]

- Injection differential pair (Q_3 and Q_4)
  - Generate 3rd harmonics
- Differential Colpitts oscillator (Q_1 and Q_2)
  - Free running frequency ~ $3 \cdot f_{in}$
- Narrow injection locking range
- Low power efficiency

Triple Push Frequency Tripler

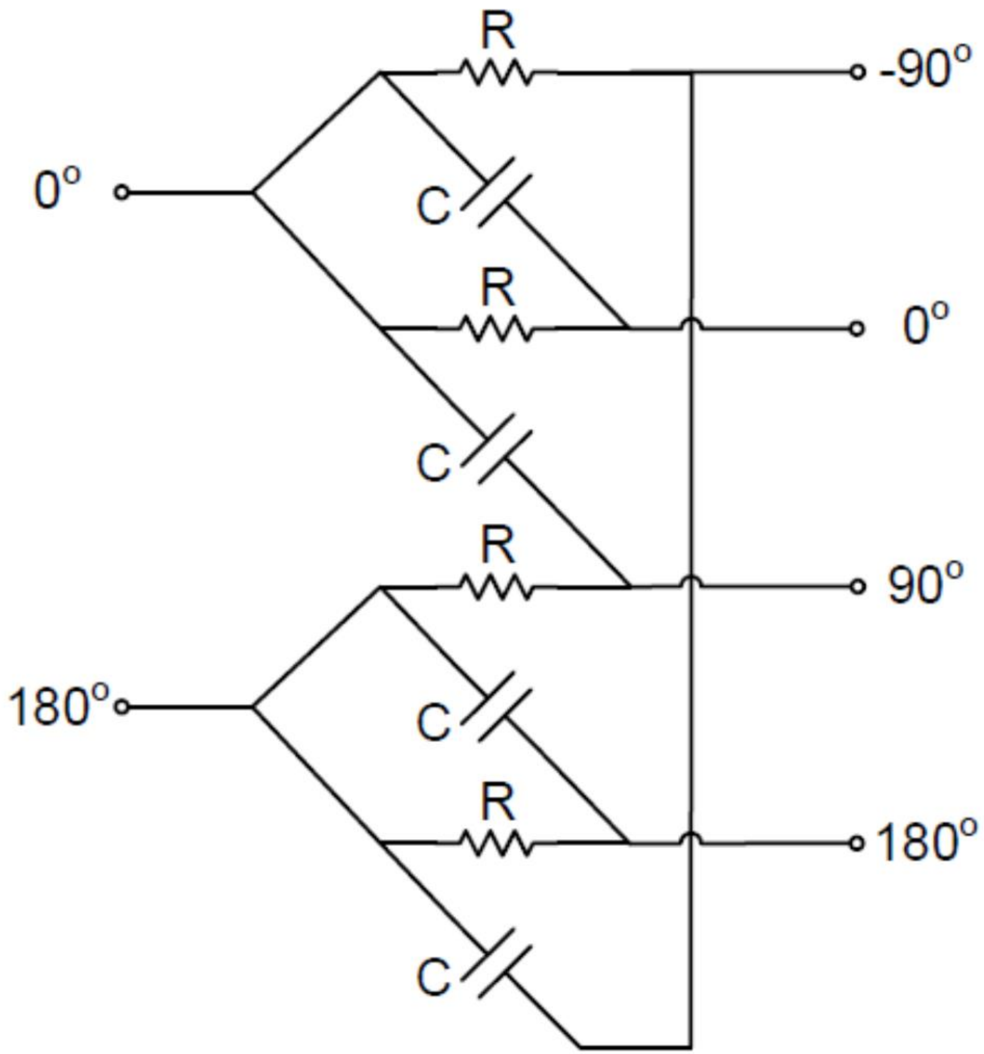


- An M-stage ring oscillator provide M signals with a $360^{o}/M$ phase difference in the fundamental frequency f_0
- The in-phase constructive combination of the M signals yield an output frequency of $M{\cdot}f_0$



Ref [9]

Phase Shifter – Quadrature Phase Generation



- Quadrature phase generation
  - Quadrature VCO
  - VCO at 2x frequency followed by div-2
  - Polyphase filter
  - Injection locked ring oscillator
  - Quadrature hybrid
- Polyphase filter
  - At RC 3dB frequency, it generate quadrature output from differential input
  - Narrow band frequency response
  - Multiple stages can be used to widen the bandwidth at the cost of insertion loss
  - Small area

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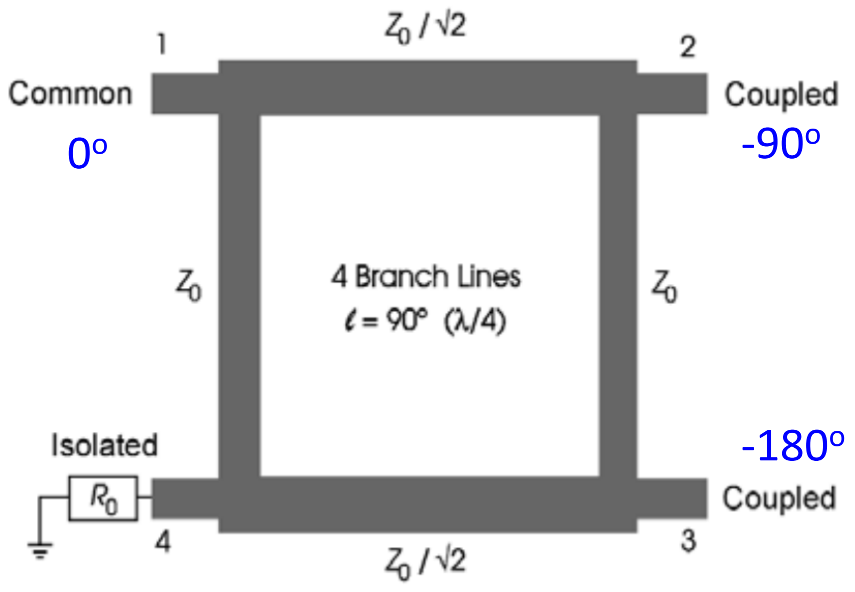
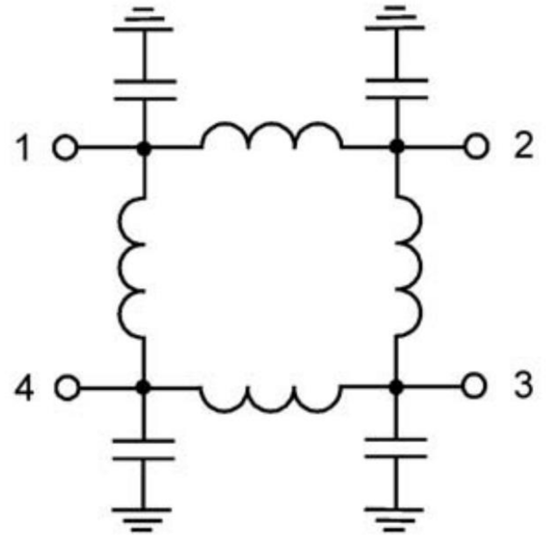
Phase Shifter – Quadrature Phase Generation

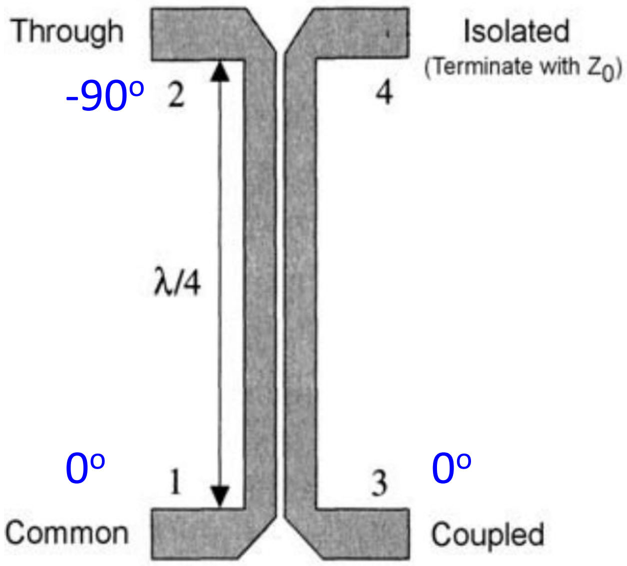


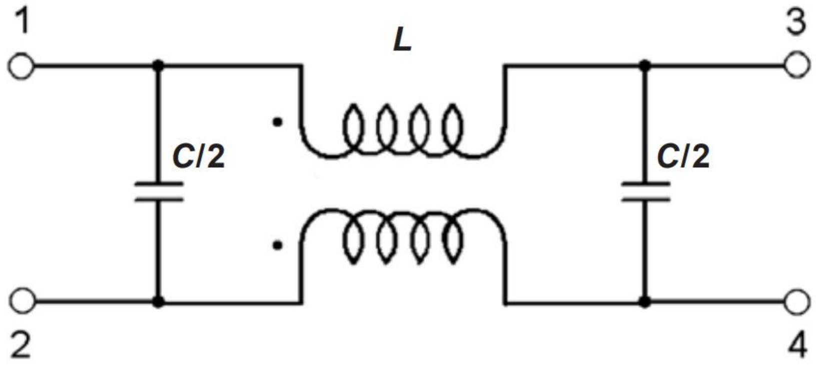


Ref [11]

- Branch line quadrature hybrid
  - Input port 1, output ports 2 and 3, isolated port 4
  - At the designed frequency, power is delivered from port 1 to port 2 and 3 with equal power and 90^o phase difference
  - No power loss at port 4
  - Transmission line implementation and lumped element implementation

Phase Shifter – Quadrature Phase Generation



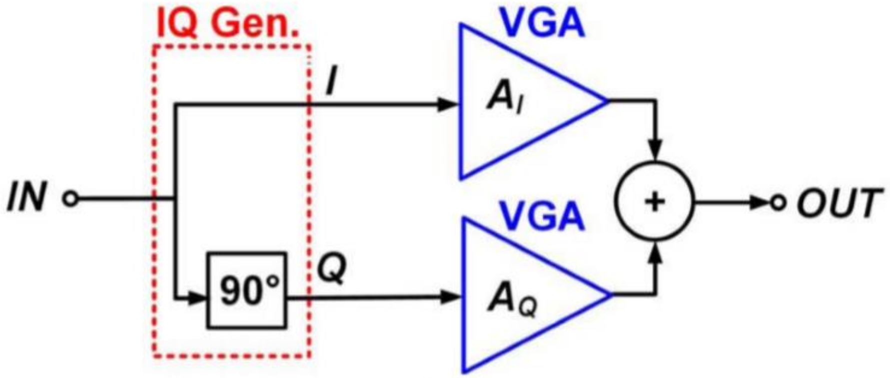


$$Z_0 = \sqrt{L/C} \qquad L = \frac{Z_0}{2\pi f_{3dB}} \qquad C = \frac{1}{2\pi f_{3dB} \cdot Z_0}$$

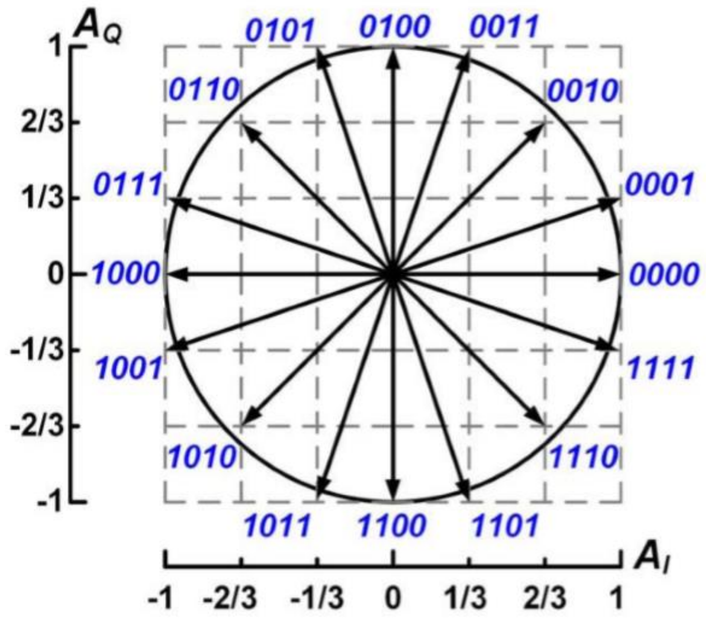
- Coupled-line quadrature hybrid
  - Input port 1, output ports 2 and 3, isolated port 4
  - At the designed frequency, power is delivered from port 1 to port 2 and 3 with equal power and 90° phase difference
  - No power loss at port 4
  - Transmission line implementation and lumped element implementation

Phase Shifter – Vector Modulator

- The vector modulator phase shifter is based on an IQ signal generator and two variable gain amplifiers (VGAs).
- The I and Q components of the input signal are weighted by properly choosing the gains A_I and A_Q of the two VGAs.
- By properly choosing the normalized gains A_I and A_Q equal to −1,−2/3,−1/3, 0, 1, 1/3, 2/3, 1, it is possible to select any phase shift from 0° to 360° with a 22.5° resolution.



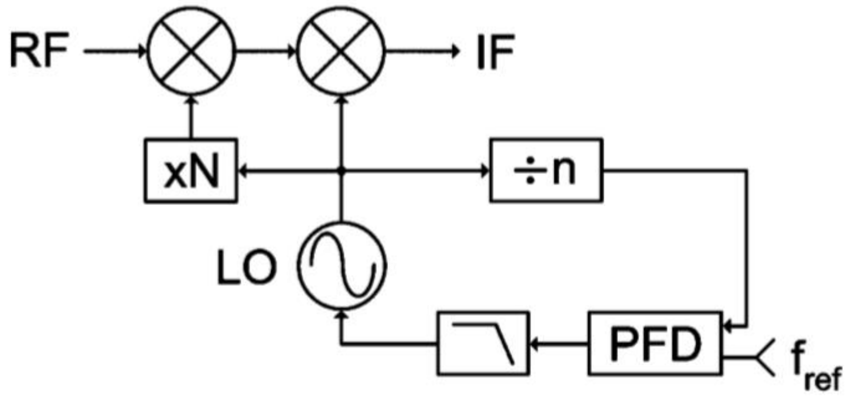
(a)

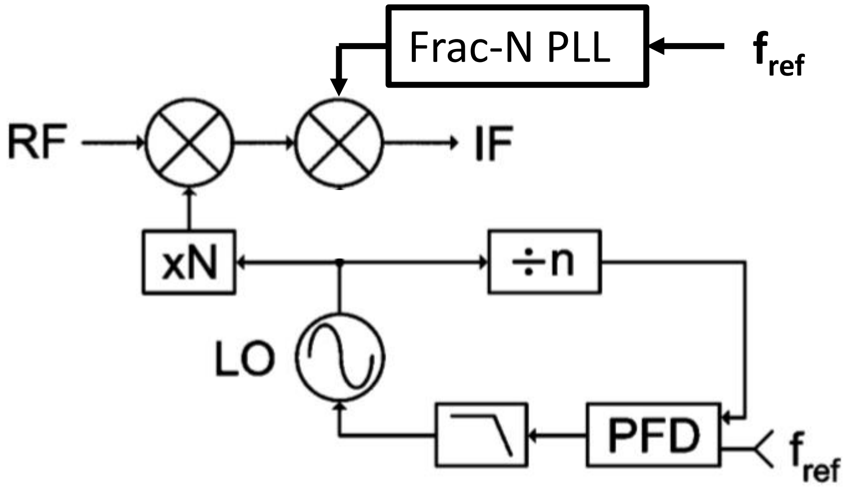


(b)

Ref [10]

Improve Phase Noise Performance at Architecture Level





Wide BW to reduce VCO PN

- Synthesizer design requirements
  - Good IPN
  - Fine resolution
  - RF frequency and IF frequency generation
- Architecture strategies
  - Fractional-N PLL for frequency resolution
  - Integer-N PLL for better PN noise performance and wide loop BW
  - High reference frequency to allow wide BW and low 20logN noise multiplication ratio
  - Reference signal generation is critical

References

[1] Kenichi Okada, CMOS millimeter-wave transceiver design, IEEE SSCS webnar, 2018

[2] Theodore Rappaport, et. al., Introduction to millimeter wave wireless communications, informit.com, 2014.

[3] K. Okada, Challenges toward millimeter-wave CMOS circuits enhanced by design techniques, IEDM 2013.

[4] Y. Zhang, et. al., Integrated Wide-Band CMOS Spectrometer Systems for Spaceborne Telescopic Sensing, TCAS-I, 2019

[5] H.-C. Park, et. al., A 39GHz-band CMOS 16-channel phase-array transceiver IC with a companion dual-stream IF transceiver IC for 5G NR base-station applications, ISSCC 2020.

[6] C. C. Wang, et. al., W-band silicon-based frequency synthesizers using injection-locked and harmonic tripler, IEEE Trans. on Microwave Theory and Techniques, 2012.

[7] E. Monaco, et. al., Injection-locked CMOS frequency doublers for u-Wave mm-Wave applications, JSSC 2010.

[8] Z. Chen, et. al., W-Band Frequency Synthesis Using a Ka-Band PLL and Two Different Frequency Triplers, RFIC 2011.

[9] C.-C. Chen, et. al., Ring-based triple-push VCOs with wide continuous tuning ranges, IEEE Trans. Microwave Theory and Techniques, 2009

[10] D. Pepe, et. al., Two mm-Wave vector modulator active phase shifters with novel IQ generator in 28nm FDSOI CMOS, JSSC 2017

[11] Gary Breed, Transmission line and lumped element quadrature couplers, High frequency electronics, Nov. 2009.